

Film Dosimetry



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Medical Physics Laboratory II

Name: Souvik Das

Roll No: 241126007



National Institute of Science Education and Research
Jatani, Odisha, India

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1 Objective

- To establish a reliable dosimetric calibration for GAFchromic™ EBT3 film by relating film response to known absorbed doses under reference irradiation conditions.
- To determine an **unknown absorbed dose** from irradiated film by converting scanner pixel values (or net optical density) into dose using the established calibration curve.
- To perform a **spoke (starshot) test** using film for evaluation of radiation isocenter behavior and beam-axis convergence as part of machine quality assurance.
- To perform a **light and radiation field congruence test** (field coincidence test) to verify geometric agreement between optical field indicators and delivered radiation field.
- To carry out **patient-specific QA** by comparing measured planar film dose with treatment planning system (TPS) dose using profile comparison and γ -index analysis.
- To understand major sources of uncertainty in film dosimetry (scanner artifacts, film handling, post-irradiation growth, lot dependence, and setup reproducibility) and apply practical methods to reduce them.

2 Apparatus

- GAFchromic™ EBT3 Film
- Teletherapy Unit (Linear Accelerator or Co-60)
- Flatbed Document Scanner (e.g., EPSON Expression 10000–13000XL / equivalent) with 48-bit RGB acquisition
- Scanner software with all automatic corrections disabled (color restoration, sharpening, dust removal, auto-exposure)
- Computer workstation with Radiochromic.com workflow and/or film analysis software (for calibration, dose map generation, profiles, and γ -analysis)
- Uncompressed TIFF image format support for archival and analysis
- Water-equivalent slab phantom (solid water) for buildup and backscatter conditions
- Ionization chamber and electrometer (used for reference dose assignment during calibration irradiation)
- Phantom positioning accessories: ruler, laser alignment, spirit level, and setup marks
- Film handling tools: powder-free gloves, guillotine/cutter, non-metallic marker for orientation marks
- Film positioning frame to keep film centered and reproducibly oriented on scanner bed
- Transparent compression sheet (glass/PMMA, 2–4 mm) for flat contact during transmission scanning
- Fiducial markers (if required) for image registration between measured and planned dose planes
- QA phantoms/fixtures for spoke (starshot) and field congruence tests

3 Theory

Radiochromic film dosimetry is widely used in modern radiotherapy because it provides high spatial resolution, near water-equivalent response, broad dynamic range, and self-developing behavior without wet chemical processing [1, 2]. For small fields, steep dose gradients, MLC-modulated deliveries, and geometric QA tests, film remains one of the most informative 2D dosimeters. The practical workflow used in this experiment follows recommended procedures from Radiochromic.com tutorials, peer-reviewed protocol papers, and the EBT3 product guidance published by Ashland [3, 4, 5, 6].

3.1 Fundamental Physical Principle

GAFchromic™ EBT3 film consists of an active microcrystalline layer between polyester substrates. Ionizing radiation initiates polymerization of sensitive monomers in the active layer, creating color centers that increase optical absorbance. The color change is dose-dependent and can be digitized by a flatbed scanner. Since EBT3 is nearly tissue equivalent, perturbation of the radiation field is small for megavoltage dosimetry [1].

For EBT3 specifically, the manufacturer describes the film as suitable for fast and highly accurate radiotherapy measurements with high-resolution images, while protocol studies recommend standardized reflection scanning and consistent film handling to reduce readout error [6, 5].

The optical density (OD) of a scanned film is

$$OD = \log_{10} \left(\frac{I_0}{I} \right), \quad (1)$$

where I_0 is transmitted light intensity through an unirradiated reference and I is transmitted intensity through the exposed film. In practice, net optical density is used:

$$\text{netOD} = \log_{10} \left(\frac{PV_{\text{unexp}}}{PV_{\text{exp}}} \right), \quad (2)$$

where PV_{unexp} and PV_{exp} are scanner pixel values under fixed scanning conditions.

3.2 Scanner-Based Readout and Image Formation

Film response is not measured directly in Gy; instead, scanner pixel values in red, green, and blue channels are acquired (typically 48-bit RGB, 16-bit/channel). Accurate dosimetry therefore depends on a stable dosimetric chain:

- film lot,
- irradiation geometry,
- scanner model and settings,
- analysis algorithm.

Important scanner effects include orientation dependence and the lateral response artifact (LRA). LRA appears as position-dependent scanner response along the lamp axis and can bias dose if calibration regions are not centered or corrected [7]. Hence, protocol-driven scanning is essential: warm-up scans, fixed orientation, centered placement, repeated scans (first discarded), and identical acquisition settings for calibration and measurement [3, 4, 5].

3.3 Calibration Theory and Dose Conversion

3.3.1 Need for Lot-Specific Calibration

Radiochromic film sensitivity varies by manufacturing lot and changes slowly with storage conditions (time, temperature, humidity). Therefore, calibration must be lot-specific and repeated periodically. Calibration doses must cover the complete clinical dose range to avoid extrapolation error [3, 1].

3.3.2 Sensitometric Relationship

Calibration establishes a mapping

$$D = f(PV_R, PV_G, PV_B) \quad \text{or} \quad D = f(\text{netOD}), \quad (3)$$

where D is absorbed dose. Modern multichannel methods use all RGB channels to separate dose-dependent signal from dose-independent perturbations (film thickness variation, scanner nonuniformity). Radiochromic.com reports use multigaussian/channel-based modeling and spline sensitometric fitting with optional lateral correction [8, 3].

3.3.3 Practical Calibration Sequence

The recommended sequence is:

1. Cut one sheet into multiple strips while preserving orientation marks.
2. Irradiate strips to known reference doses, including one unexposed strip (0 Gy).
3. Choose dose points from low dose to about 120% of maximum expected clinical dose.
4. Scan strips together at the scan center with fixed geometry.
5. Assign ROIs with homogeneous dose and fit calibration model.

Uncertainty in reference dose (usually from ion chamber traceability) propagates directly into film dose uncertainty; therefore, careful reference dosimetry is mandatory.

3.4 Determination of Unknown Dose

After calibration, the unknown irradiated film is scanned under the same protocol. Pixel values are converted to dose point-by-point through the fitted calibration function. If needed, inter-scan correction and dose-rescaling options are applied with central unexposed/known-dose reference ROIs [4].

For any ROI, the mean absorbed dose is

$$\bar{D} = \frac{1}{N} \sum_{i=1}^N D_i, \quad (4)$$

and ROI standard deviation estimates local heterogeneity plus noise:

$$\sigma_D = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (D_i - \bar{D})^2}. \quad (5)$$

3.5 Spoke (Starshot) Test: Geometric QA Principle

The spoke or starshot film test evaluates radiation isocenter size and stability. Multiple narrow beams are delivered at different gantry/collimator angles to produce a star pattern on film. Ideally, all beam axes intersect at a common point; deviations indicate mechanical/radiation misalignment.

From analyzed spoke lines, the software estimates:

- radiation isocenter coordinates,
- isocenter radius (or diameter),
- displacement relative to fiducial/laser marker.

A smaller reconstructed isocenter radius indicates better geometric precision of treatment delivery, which is critical for IMRT/VMAT and stereotactic techniques.

3.6 Field Congruence Test (Light vs Radiation Field)

Field congruence verifies whether the light field projected by the treatment unit coincides with the actual radiation field delivered to the film. In QA terms, this checks setup fidelity between optical positioning and dose delivery.

Profiles across field edges are analyzed to locate radiation boundaries (often by 50% dose level/FWHM for dose maps, or inflection-point approximation for raw images). Differences between light-field marks and radiation boundaries are reported as directional displacements on each semiaxis. Significant disagreement can cause geometric targeting errors and requires machine adjustment.

3.7 Patient-Specific QA with Film

For patient-specific QA, measured planar dose on film is compared with TPS-predicted planar dose for the same geometry and depth. The comparison includes visual overlays, profile analysis, dose difference maps, and gamma analysis.

The 2D gamma index combines dose-difference and distance-to-agreement criteria:

$$\gamma(\mathbf{r}) = \min_{\mathbf{r}_r} \sqrt{\left(\frac{\Delta D(\mathbf{r}, \mathbf{r}_r)}{\Delta D_{\text{crit}}}\right)^2 + \left(\frac{\Delta r(\mathbf{r}, \mathbf{r}_r)}{\Delta r_{\text{crit}}}\right)^2}. \quad (6)$$

Points with $\gamma \leq 1$ satisfy the acceptance criterion. Common clinical criteria are 3%/3 mm or 2%/2 mm with a low-dose threshold (e.g., 10%). The gamma passing rate is

$$\% \text{ GP} = \frac{N_{\gamma \leq 1}}{N_{\text{eval}}} \times 100. \quad (7)$$

High passing rates indicate agreement between planned and delivered dose distributions; low passing rates prompt investigation of setup, TPS modeling, MLC delivery, output, or film workflow.

3.8 Sources of Uncertainty and Their Control

Major uncertainty components in film dosimetry are:

- **Film related:** lot variability, handling marks, orientation inconsistency, post-irradiation growth.
- **Irradiation related:** setup reproducibility, phantom alignment, reference-dose uncertainty.
- **Scanning related:** warm-up instability, LRA, inter-scan variability, film curling, nonuniform contact.
- **Analysis related:** ROI selection bias, registration error, model fitting uncertainty, inappropriate extrapolation.

Control measures include strict orientation marking, fixed post-irradiation scanning time (typically around 24 h), centralized ROIs, repeated scans with first-scan rejection, multichannel calibration, and consistent correction strategy [1, 3, 4].

3.9 Clinical Relevance

The combined use of calibration-based unknown dose determination, spoke test, field congruence test, and patient-specific QA provides both **dosimetric** and **geometric** verification in radiotherapy. Film dosimetry therefore acts as an end-to-end QA tool linking machine performance, treatment planning accuracy, and delivery fidelity, ultimately supporting safer and more accurate patient treatments.

4 Observation

5 Conclusion

A comprehensive understanding of the governing physical principles and stringent Quality Assurance protocols ensures optimal clinical outcomes and fundamental radiation safety.

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